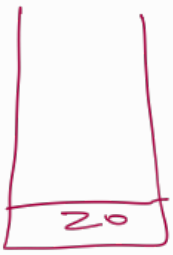
Pda recognises cfg



Stack alphabet

$(Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

- Q: finite set of states
- Σ: input alphabet
- δ: transition fn
- q_0: initial state
- Z_0: starting stack symbol
- F: set of final state

$$\delta = Q \times (\Sigma \text{ union } \{\phi\}) \times \Gamma \rightarrow Q \times \Gamma^*$$

PDA for $(a^n b^n)$ ab, aabb, aaabbb.....

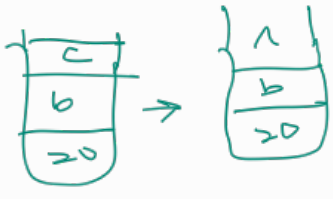
Push

$a, Z0 \mid aZ0$

- a: input
- Z0: top of stack
- aZ0: top two elements

Pop

$a, C \mid \wedge$



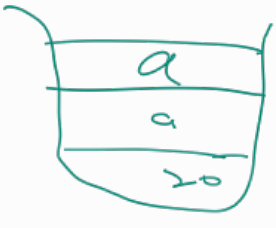
Skip:: $a, Z0 \mid Z0$ with consuming input

Skip without consuming input $\wedge, X \mid Xa$



→ push all a's then for every a pop b.

a	a	a	b	b	b	^





reaching string last

$\delta(q_0, a, z_0) = (q_0, az_0)$ — top

- q_0: current state
- a: input
- az_0: a is pushed.

$b, a / \wedge$

$\delta(q_0, b, a) = (q_1, \wedge)$

poping

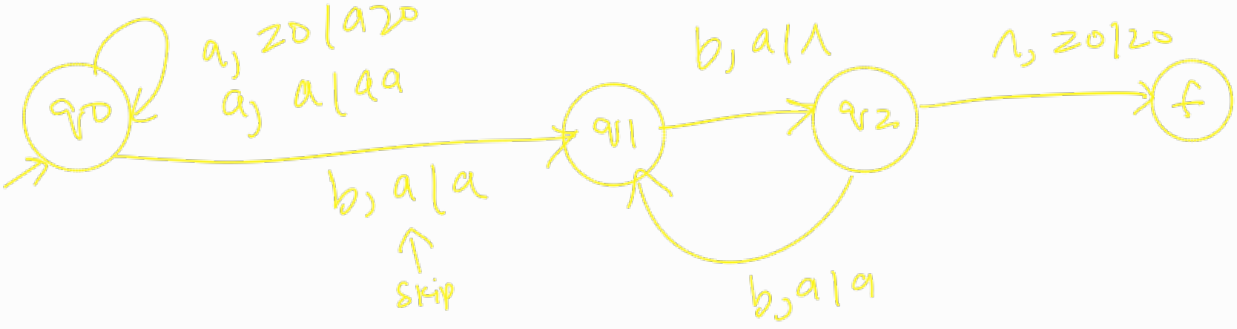
$a^n b^{2n}$

push -2a's pop every 1 b.

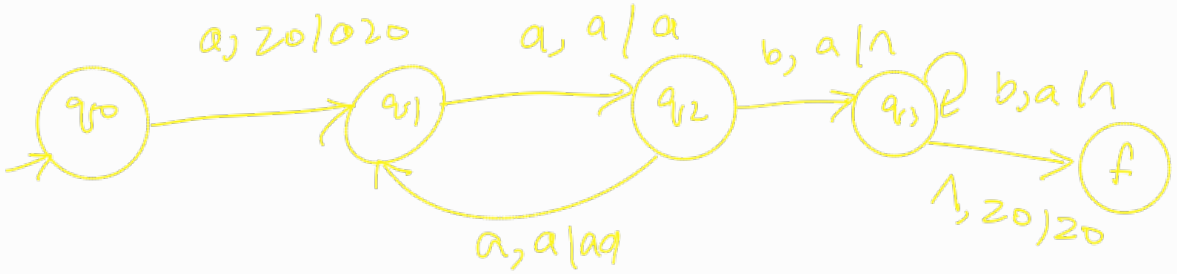
a	a	b	b	b	b	∧

aaaa bbbbbbbb ∧

Push all a's.
Skip every first b then
pop one b corresponding to it.



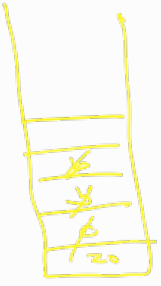
$a^{2n} b^n$ → push first a to stack, Skip one a, repeat till one a's are first pop one b's.

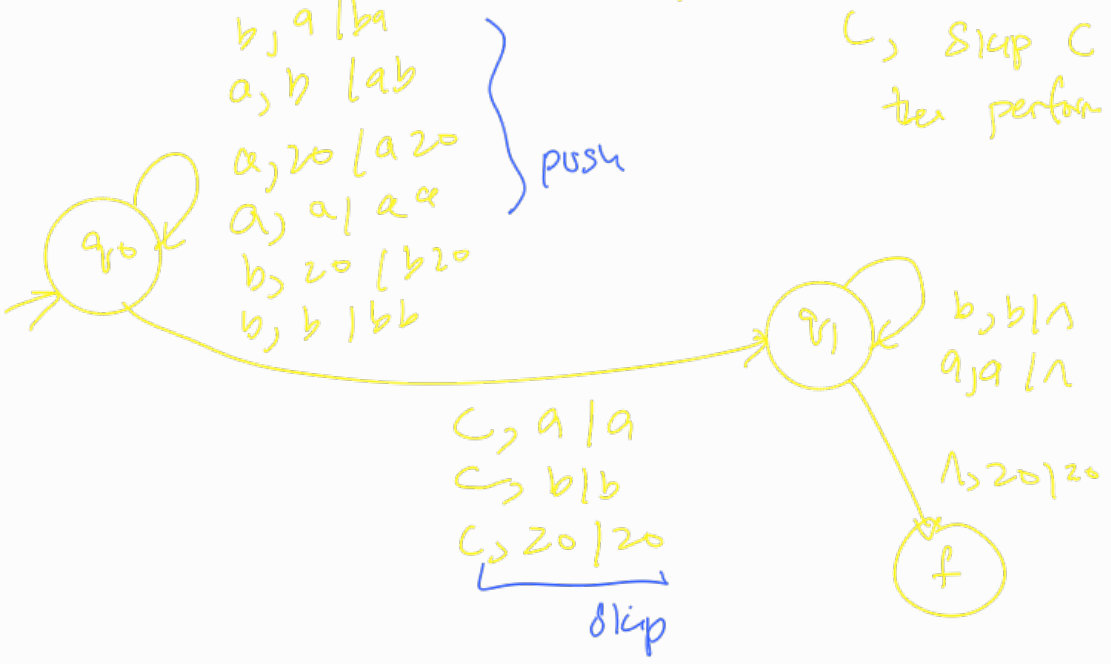


PDA for wcw^r

a	b	b	c	b	b	a	ʌ

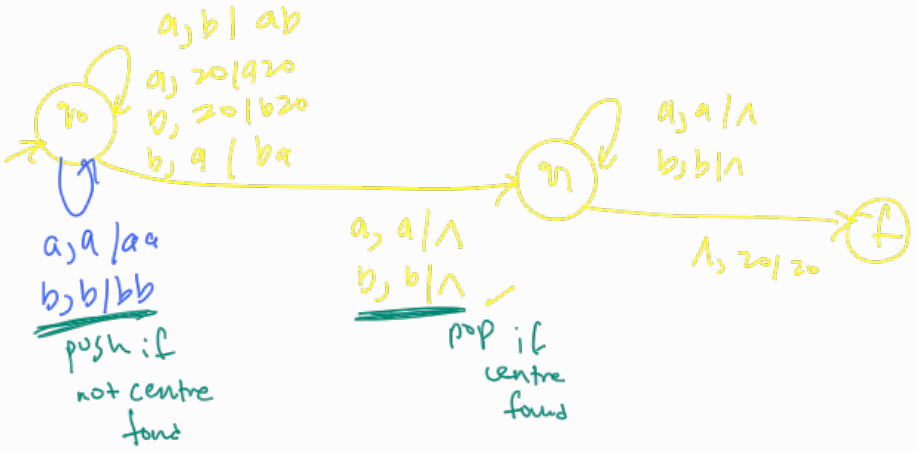
Push all symbols till we get c, Skip c and then perform pop.





e.g. ww^R NPDA → a b a b b a b a



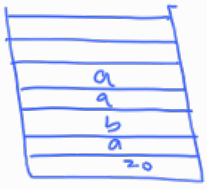


ww^R

a a a a a a

b b b b b b

↓ no centre so just push.



aaa aaa

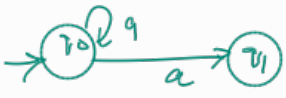
if top of stack element is same as input

push → pop due to aaa|aaa
aba|aba

DPDA



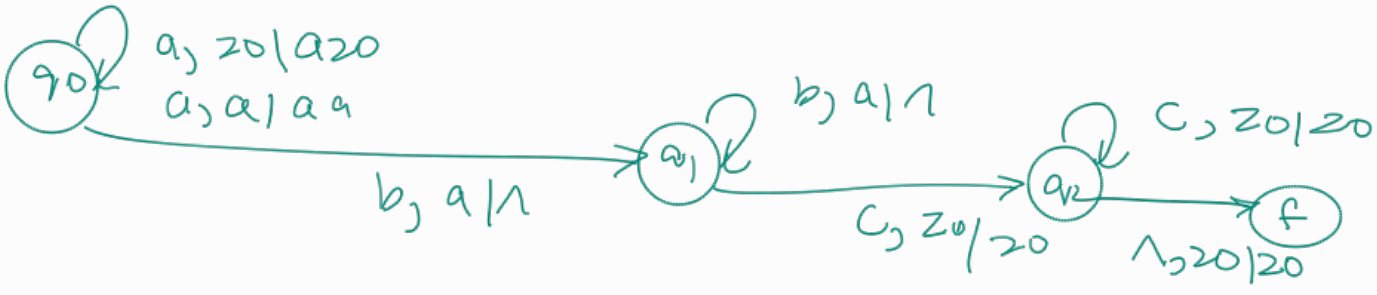
NPDA



$a^i b^j c^k$ $i=j$ $i, j, k >= 1$

abccc ... aabbc ...

for every a pushed pop b and if input symbol is c skip



$a^i b^j c^{k+l} d^l$ $\quad i==k$ or $3j == l$ or $j == 2k$ $\quad \geq 0$

atleast 1 condition should be accepted.

lets say $2j = 3l$

make any subject

$$l = \frac{2j}{3}$$

$l = 3j$

$j = 1$

$l = 3$

So it means we have more d's then b's.

$j \neq 1$ $\quad j \neq 2$ $\quad j = 3$ ✓

$l = \frac{2(3)}{3} = 2$

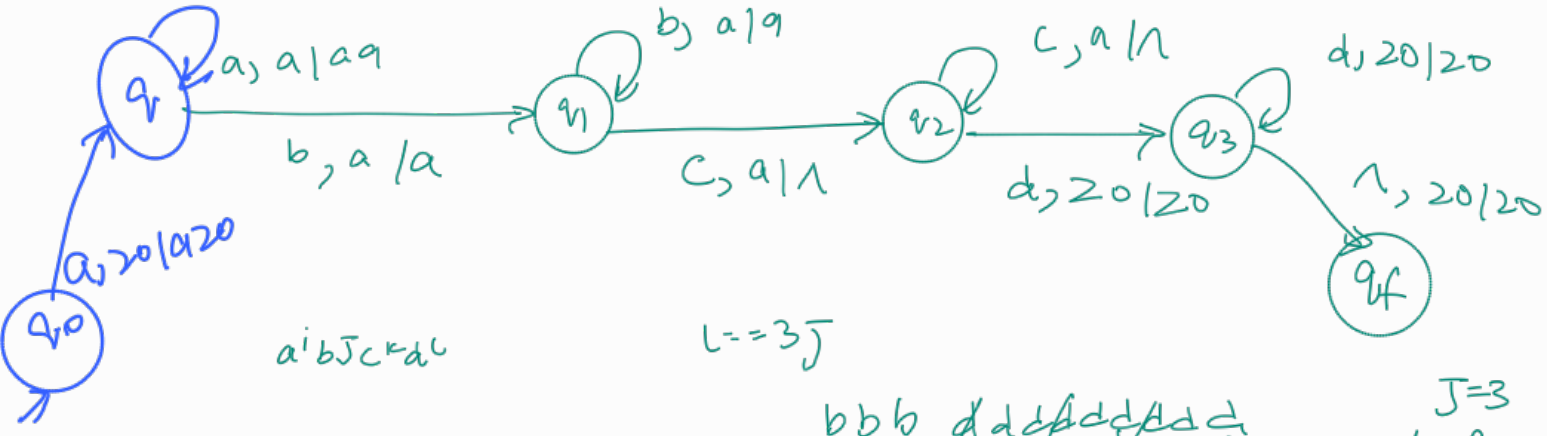
So min possible b's are 3 where d's will be 2.

$i == k$ or $l == 3j$ & $j == 2k$ either satisfy Both.

1) $\wedge$
2) aabccd
3) aaabbcddddd
4) abbcd
5) aabccddd
6) aabbcddddd
7) aaaabccccd
8) abbbbcddddddddddd
9) abbbbbbcccd
10) abcddd

aaabccd

i = k



$a^i b^J c^k d^L$

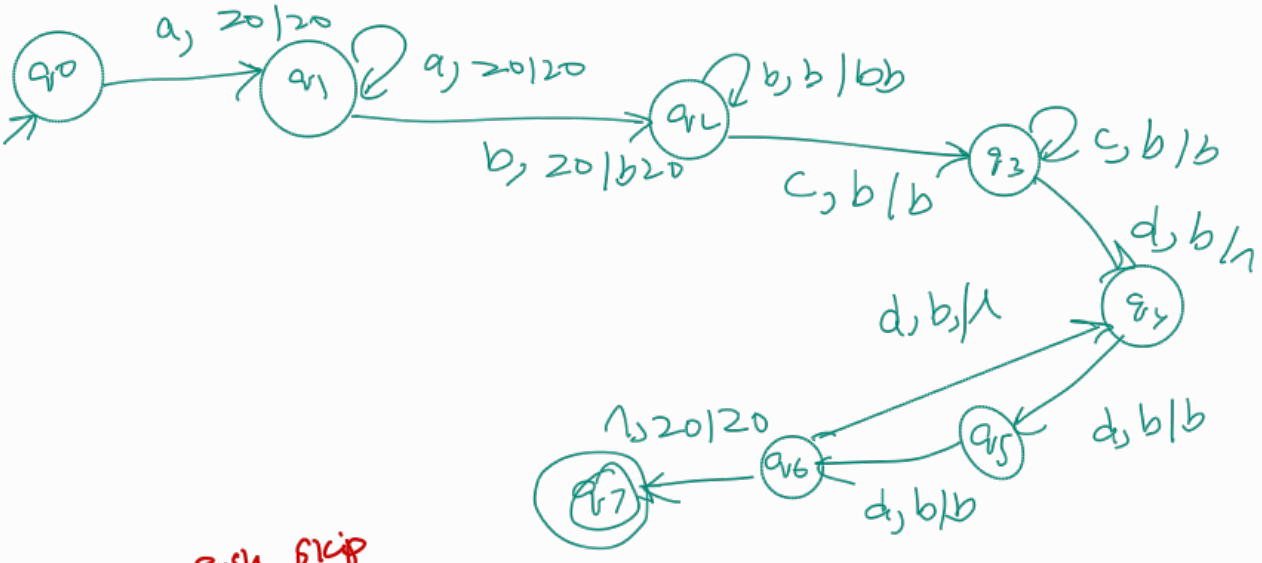
$L = 3J$

bbb ddddddddd

J=3
L=9

aa bbc ddddddd

push all b's
so pop 2d and skip 2d's.



push skip

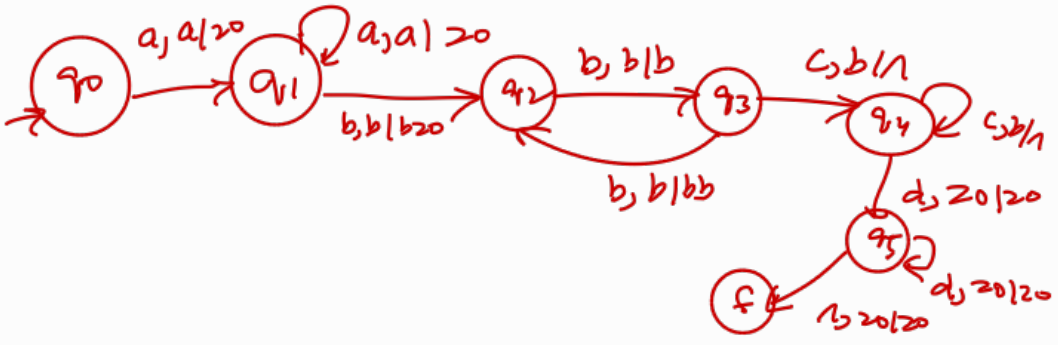
b b b b b b b b

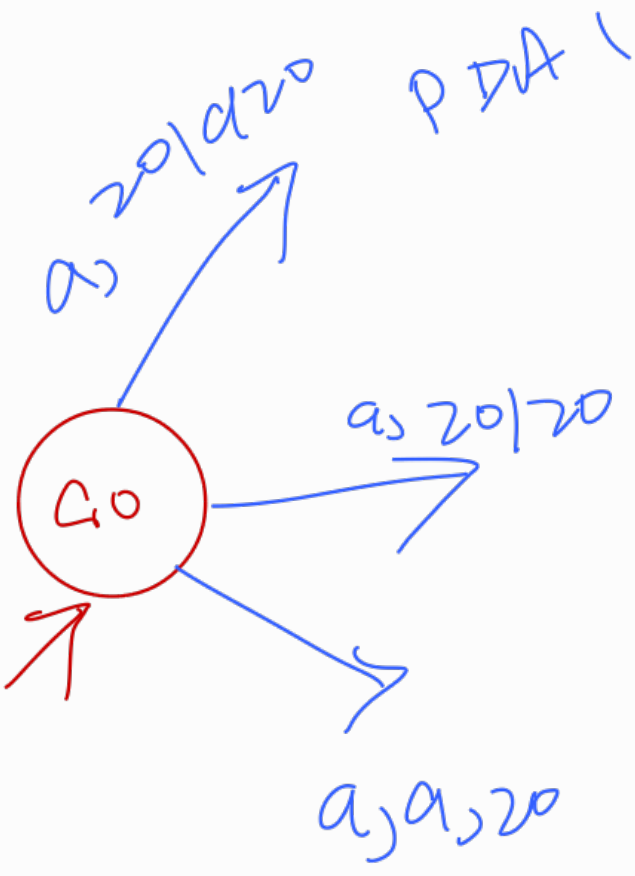
cc

ccc

$a^i b^J c^k d^L$

aa bbbb cc d




PDA
q_0
$q_1 z_0 | a z_0$
$q_1 z_0 | z_0$
$q_1 q_1 z_0$

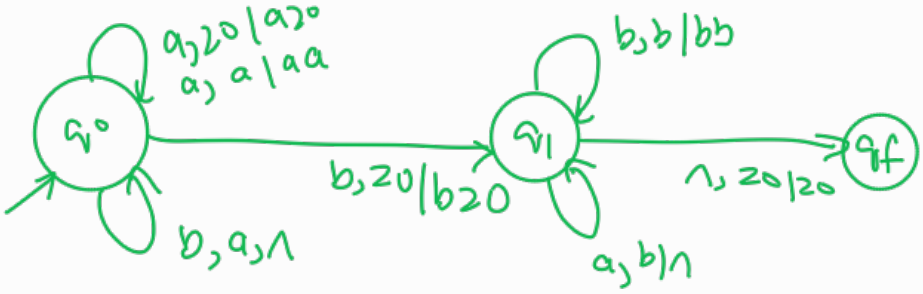
$\{a^n b^{m+n} a^m \; n,m \geq 0\}$

a bbb aa ∧
aa bbbb aa ∧
aaa bbbbb a ∧

no. of as = no of bs

$a^n b^n b^m a^m$
aabb bbb aaa
aa bb bb aa

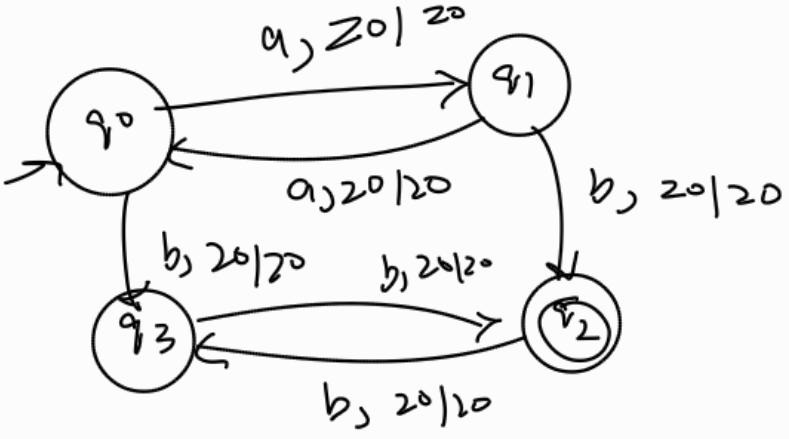




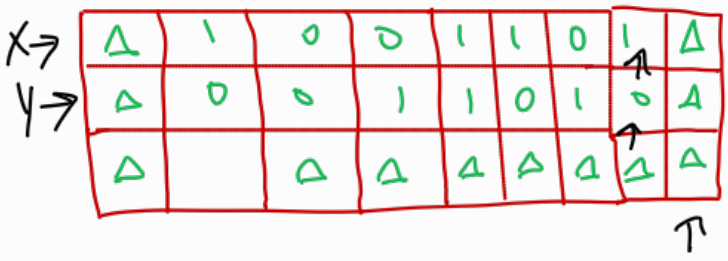
$L = \{ a^n b^m \quad n+m = \text{even} \}$

aabb
aaaa
a bbb

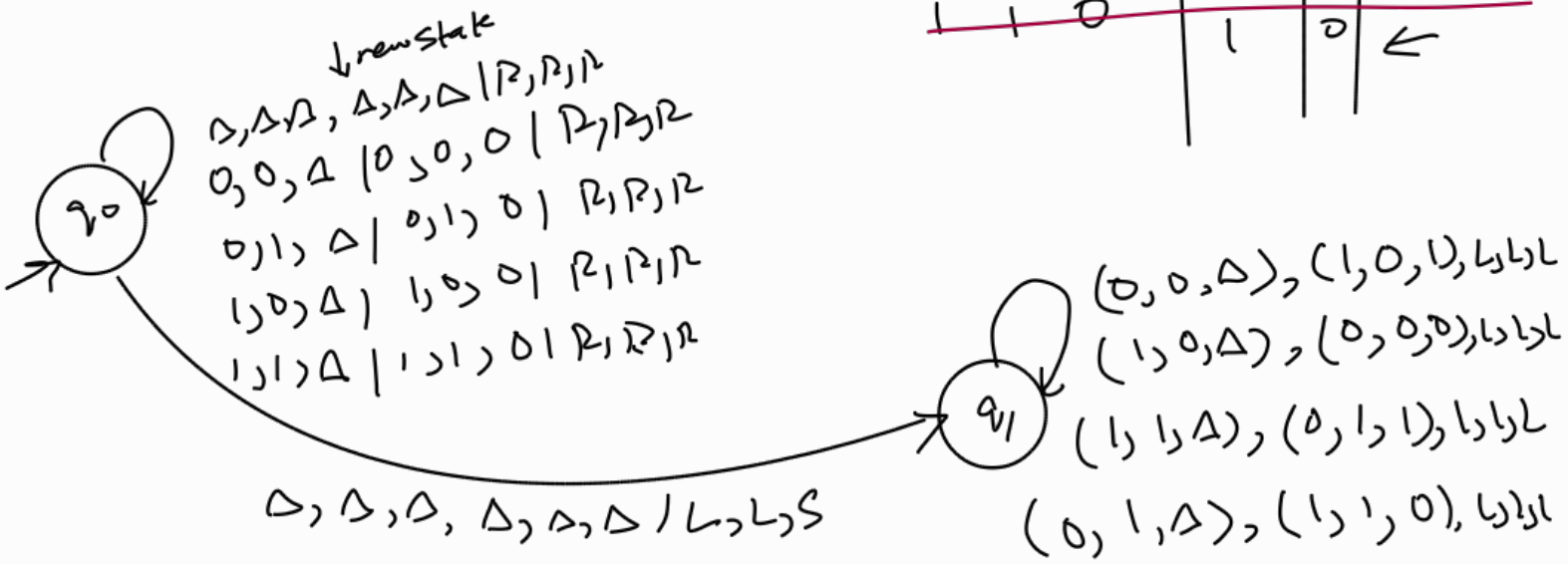
aaa aaa bb





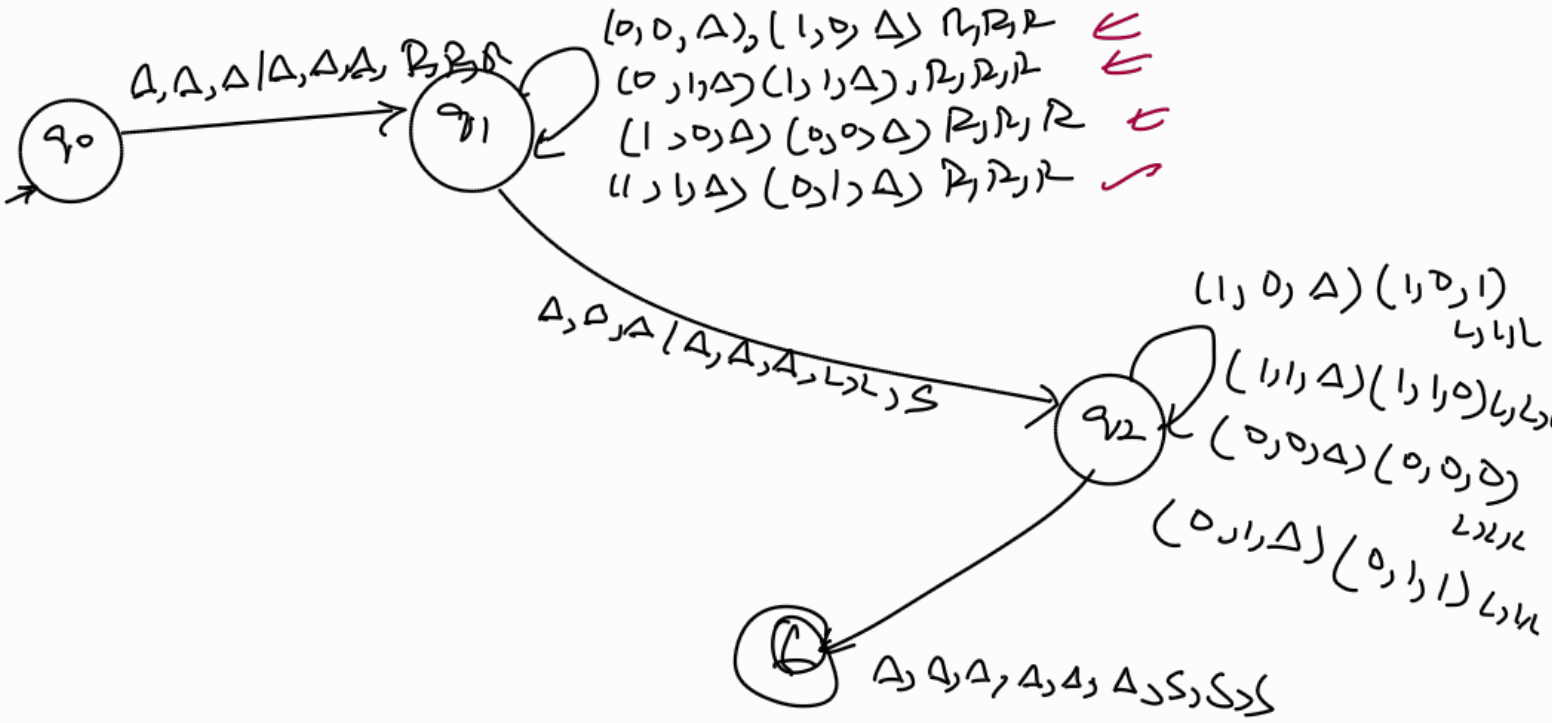


x	y	$\bar{x}$	$\bar{x}+y$	c
0	0	1	1	0
0	1	1	0	1
1	0	0	0	0
1	1	0	1	0



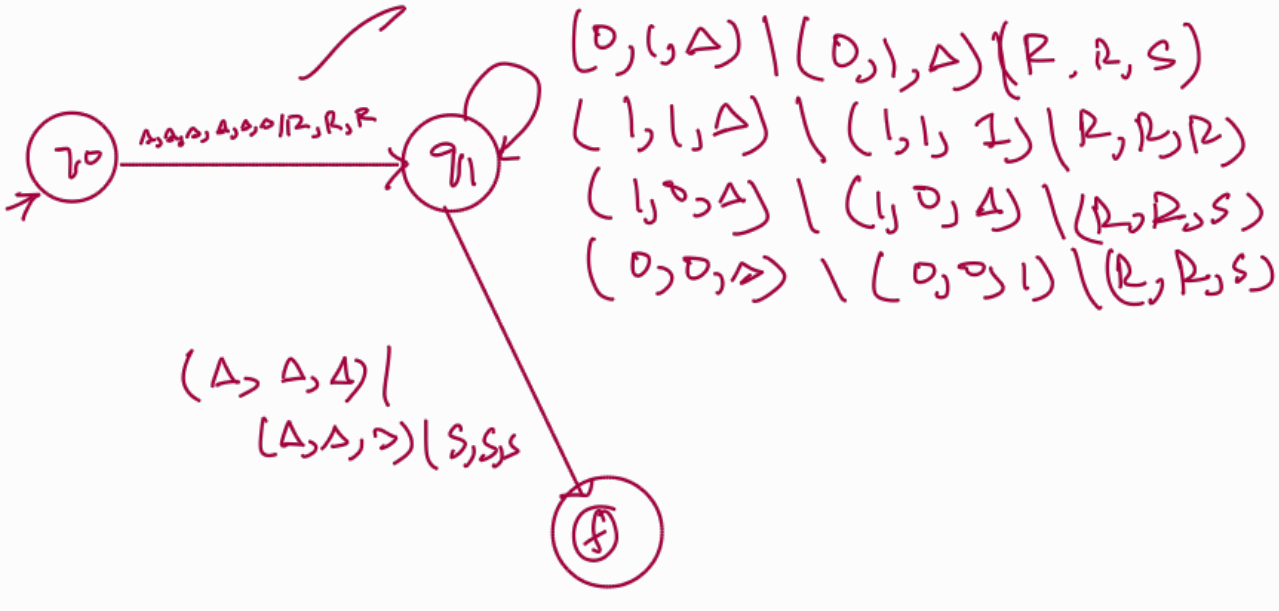
$\bar{x}$	y	$\bar{x}+y$	C
1	0	1	0
1	1	0	1
0	0	0	1
0	1	1	0

$\bar{x}+y$



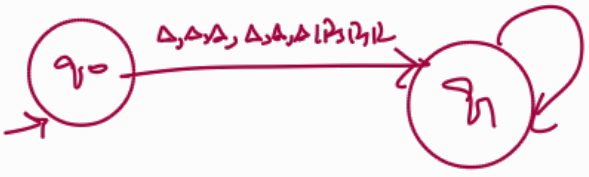
T1 point till Δ before that

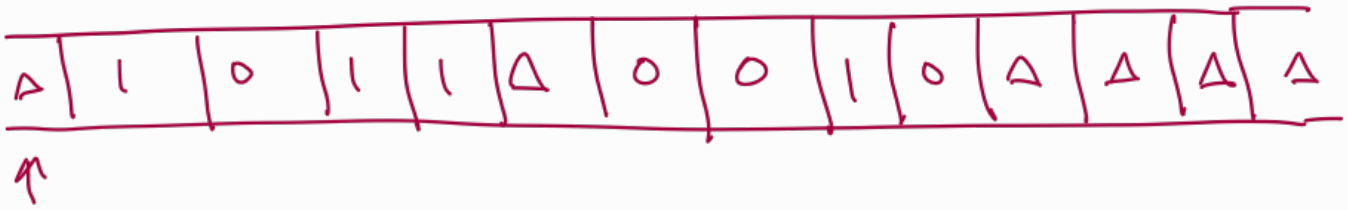
XNOR
Same = 1
diff = 0



T1	Δ	0	1	1	1	1	0	0	Δ
T2	Δ	1	0	1	Δ	Δ	Δ	Δ	Δ
T3	Δ								Δ

Хотя
Сигма: 1
dif = 0





Δ 1 0 1 1 Q 0 0 1 0 Δ Δ Δ ...

Δ Q 1 1 1 0 Δ

Δ Q A 1 1 0 Δ

Δ Q A 1 1 0 Δ X

Δ Q A A 1 0 Δ X X

Q A A A 0 Δ X X X

Q A A A B Δ X X X

Q A A A B Δ X X X X

1 0 1 1 Δ 1 1 1 0 Δ X X X X X

D A 1 1 1 0 Δ

B Δ X X X X X

1 0 1 D Δ 1 1 1 B Δ X X X X 1 Δ

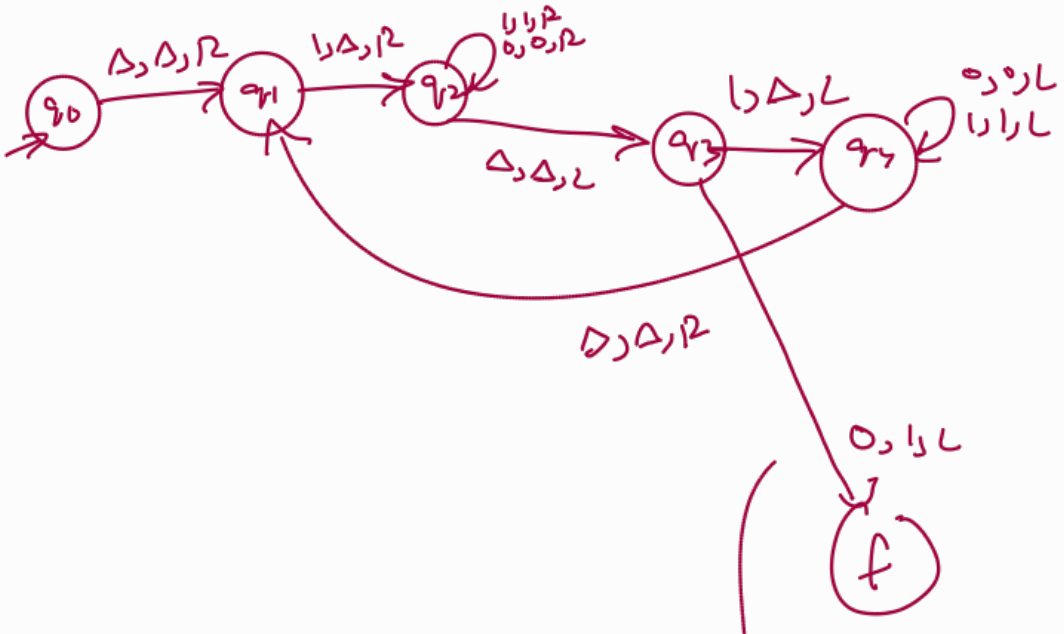
D D Δ 1 1 1 B Δ X X X X 1 Δ

TM for Subtraction of 2 unary Integers

1111 − 11 = 11

11 − 111 = 0 (X)





logic for first 1 mark it Δ.

→ and for this mark the last most 1 with Δ.



↑ Replaced 0 by 1 because

1111 − 11 → (11)

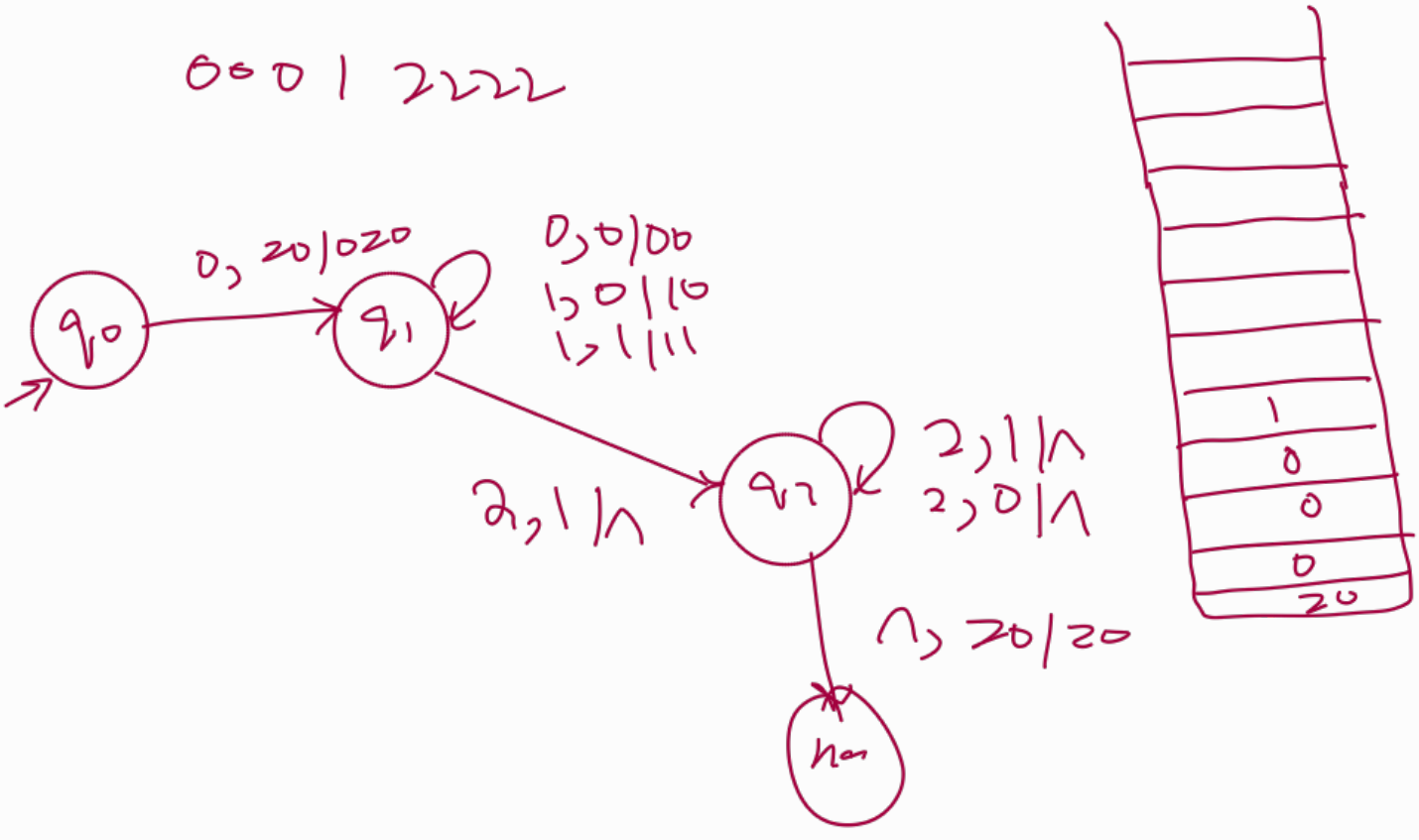
$L = \{0^n 1^m 2^{(n+m)} \mid m, n \geq 0\}$

n=2
m=3

n=3
m=1

00 111 22222

0 0 0 1 2222



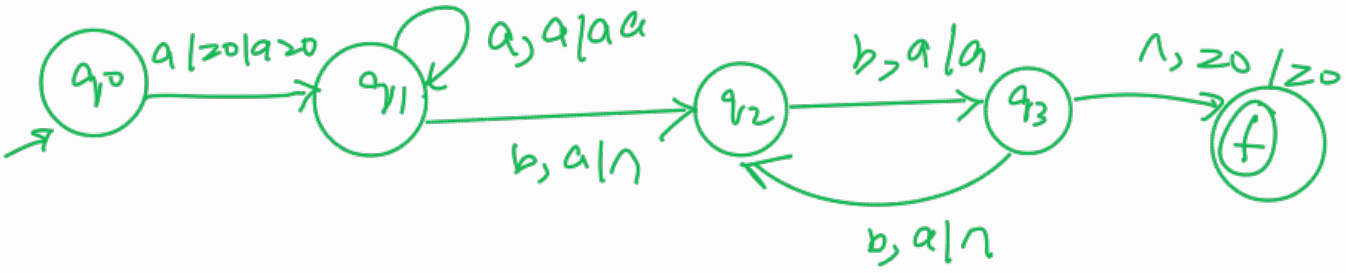
$a^m b^{(2m+1)}$ $m \geq 1$

m=4

aaaa bbbbbbbbb

1 more

dPDA



$a^m b^{2m+1}$

m=3

2(3)+1

7

aaabbbbbbb∧

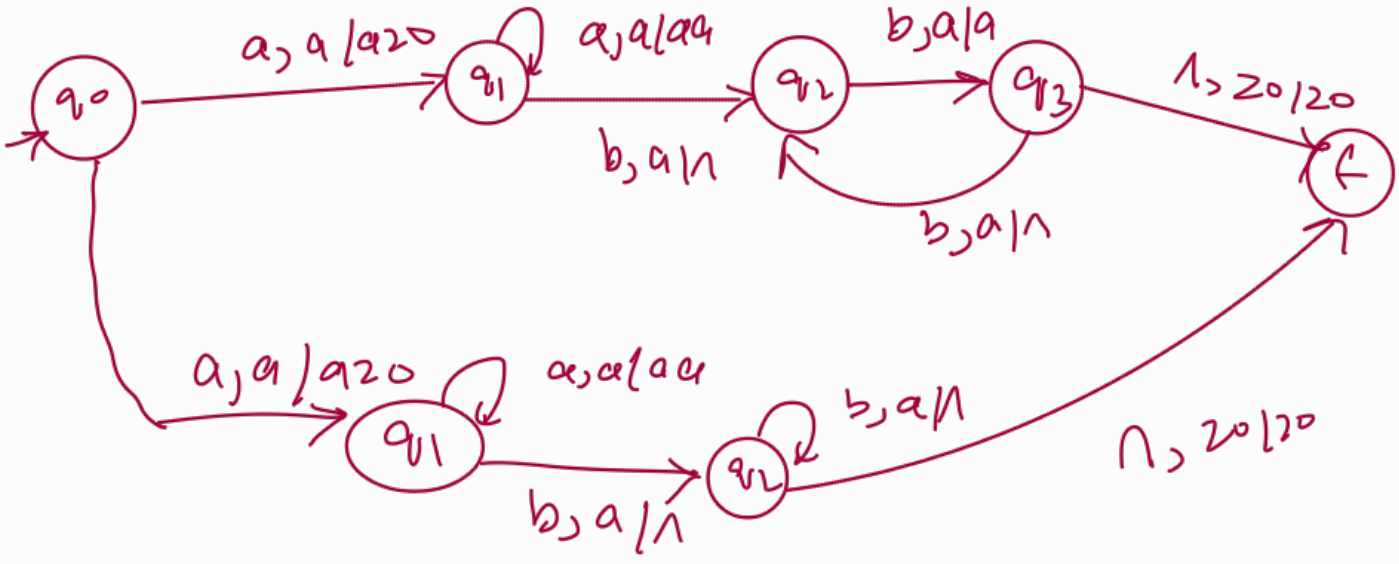
$a^n b^{2n}$ $n \geq 1$ ∪ $a^n b^n$ $n \geq 1$

$n-2$

aabbbb

aabb

18, 19



equal not a's and b's

NPDA → 1 final state

Pumping lemma → used to prove that a language is not context free

let L be CFL. let n be integer constant.

→ Select a string 'z' from 'L' such that $|z| \geq n$.

Divide the string into 5 parts. uvwxy such that

i) $|vwx| \leq n$

ii) $|vx| \geq 1$

for $i \geq 0$, $uv^i w x^i y$ is in L. $i = 0, 1, \ldots$

Show that $L = \{a^n b^n c^n \mid n \geq 1\}$ is not CFL.

1) let L be CFL

2) Assume let $n = 3$

3) selecting a string (z) abc aabbcc aaabbbccc

So $z = aabbcc$

$|z| \geq 3$

4) Dividing string into 5 parts. $\underbrace{aa}_{u}\ \underbrace{a}_{v}\underbrace{b}_{w}\underbrace{b}_{x}\ \underbrace{bCC}_{y}$

$|vwx| \leq n$

i) $|abb| \leq 3$ ✓

ii) $|ab| \geq 1$ ✓

$u = aa$ $x = b$

$v = a$

$w = b$ $y = bccc$

$uv^i wx^i y$

$i=0$ $aaa^0 bb^0 bccc \rightarrow aabbccc$ ✗ not present in Language

$i=1$ $aaabbbccc$

hence not CFL.

$\longleftrightarrow$

$L = \{ a^p \mid p \text{ is prime no} \}$ $p = 2, 3, 5, 7$

aa
aaa
$aaaaa$

$L \rightarrow CFL$
let $n=4$
$z = aaaaa$

$|aaaaa| \geq 4$ ✓

Dividing into 5 parts

$u\ v\ w\ x\ y$

$aa\ aaa$
$u\ v\ w\ x\ y$

$|aaa| \leq 4$ ✓
$|aa| \geq 1$ ✓

Pumping

$uv^i wx^i y$

$i=0$ $aa^0 a\ a^0 a \rightarrow aaa$

$i=1$ $aa^1 a\ a^1 a \rightarrow aaaaa$

$i=2$ $aa^2 a\ a^2 a \rightarrow$

$aaaa\ aaa$ ✗

$i=3$

$aaaa\ a\ aaa\ a \rightarrow$? ✗

Not a CFL

$\{a^{n^2} \mid n \geq 1\}$ Prove not CFL using pumping lemma

→ aa, aaaa, aaaaaa

$n = n$

$2 = 6$ aaaaaa $|z| \geq n$

P.L → Prove not Regular languages

P.L → Prove not CFL (P PA)

P.L 1 To prove that a lang is not Regular ↳ Regular

String = uvw ↓ non-empty (Repitation) $xy^i z$

$uv <= n$ $n \rightarrow$ length of String.

$S = a^n b^n$

$n = uvw$

$|w| \geq p \rightarrow$ length of string $L = \{w\}$

$w = xyz$ divide string in

- $|y| > 0$
- $|xy| \leq p$
- $xy^k z$ $k \geq 0$

if L is Regular it satisfies pumping lemma

(Pumping lemma - 1)

Suppose $\{0^n 1^n \mid n \geq 0\}$

i) Suppose L is regular

ii) Suppose P → pumping constant.

P → any no.

iii) chose any w.

$w = 0^P 1^P$

iv) Breaking w → String.

into xyz

the first two pieces of string are atmost P.

So x and y are inside 0^P.

← i) $|xy| \leq P$

ii) $|y| \geq 1$

v) Chose ai such that

$xy^i z \notin L$

Breaking every possible way to breaking w (String)

xyz

→ $|xy| \leq P$

→ y cannot be null

=> $x = 0^k$, $y = 0^q$, $k \geq 0$, $q \geq 1$

$k + q \leq P$

k → no of zeros in x

q → no of zeros in y excluding Λ.

$z = 0^{P-k-q} 1^P$ (starting 0's)

↳ rest of string + leftover zeros

Chose i S.t

$xy^i z \notin L$

copy paste

$0^k 0^{iq} 0^{P-k-q} 1^P$

$0^{k+iq+P-k-q} 1^P$

$0^{P+qi-q} 1^P$

$P + qi - q \neq P$

$iq - q \neq 0$

$i \neq 1$ if $q \neq 0$

(1 ≠ -1)

Exponents = Exponents

hence proved not Regular

$0^{2p}1^p$ $\quad xyz$

$|xy| \leq p$ $\quad x = 0^k$

$|y| \geq 1$ $\quad y = 0^q \quad q \geq 1$

$z = 0^{2p-k-q}1^p$

$xy^iz = 0^k 0^{iq} 0^{2p-k-q} 1^p$

$= 0^{2p+iq-q} 1^p$

171

$0^n 1^m : n < 3m$

$w = 0^{3p-1}1^p$ String should match the condition

$x = 0^k$

$y = 0^q$

$z = 0^{3p-1-k-q}1^p$

$0^k 0^{iq} 0^{3p-1-k-q} 1^p \neq p$

$0^{3p-1+iq-q} 1^p$

Pumping $\hookrightarrow$ proof

P.L for CFL. (not a CFL)

→ let L be a CFL

→ exist a pumping constant p for L.

$w = uvxyz$

1) $|vy| \geq 1$

2) $|vxy| \leq p$

3) find i such that $uv^i xy^i z \notin L$

$a^n b^n c^n : n \geq 0$

Assume L = CFL

$w = a^p b^p c^p$

Look at all decompositions into $uvxyz$ (according to Rules)

1) vy contains a's Lets say i pump up to i = 2.

$uv^2 xy^2 z$ has more a's than b's or c's

Hence $a^n b^n c^n$ doesn't hold.

(ii) v and y contains only b's or contains c's.

$\therefore$ again a's will be less. $\notin L$

(iii) v and y contains only a's and b's

no of c's becomes less

hence $\notin L$

Hence not CFL